MIL-T-28735(EC)
AMENDMENT 6
23 October 1978
SUPERSEDING
AMENDMENT 5
20 September 1974

MILITARY SPECIFICATION

TEST SET, RADAR, AM/UPM-137

This amendment forms a part of Military Specification MIL-T-28735(EC) dated 3 April 1970.

Page 1

Title: Delete and substitute: "TEST SET, RADAR, AN/UPM-137A".

Page 2

2.1, STANDARDS, MILITARY: After "MIL-STD-167" add "MIL-STD-454 Standard General Requirements for Electronic Equipment ".

Page 4

3.5.7: Delete "drip proof" and substitute "dripproof".

Page 6

3.6.16: Delete.

3.6.18.2, first sentence: After "radiation" delete "." and add "in accordance with the Microwave and rf radiation and X radiation paragraphs of MIL-STD-454, Requirement 1."

3.6.20, line 1: Delete "approvel" and substitute "approval".

3.6.20.1, line 2: Delete "inadvertant" and substitute "inadvertent".

3.6.21, line two: Insert "without removal of or damage to adjacent modules" between "case" and "and".

Page 7

3.7.2.1, last sentence: Delete.

Page 8

3.7.2.3, (b), Width column, lines 2 and 4: Delete "0.3" and substitute "0.2".

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Page 9

3.7.2.4.4 Delete in its entirety and substitute:

"3.7.2.4.4 Pulse width. The width of each pulse in the train shall be simultaneously variable over a continuous range of less than 0.2 to greater than 1.0 µsec by means of a single front panel control. Nominal pulse width shall be 0.45 µsec."

3.7.2.4.8, first sentence: Delete.

Page 10

3.7.2.5.1 Delete in its entirety and substitute:

"3.7.2.5.1 Trigger. The SIF Coder Unit No. 2 shall be capable of being triggered by either an external trigger or by the internal trigger which also triggers coder Unit No. 1. Choice of trigger source shall be made by a front panel switch which may be concentric with the pulse width adjust for coder unit No. 2. For internal triggering, an independent delay shall allow continuous train to train (first framing pulse of coder unit No. 2, relative to the first framing pulse of coder unit No. 1) displacement over the range of less than 0.5 µsec to at least 200 µsec. This delay shall also function when coder unit No. 2 is triggered externally".

Add new paragraph as follows:

"3.7.2.5.1.1 External trigger (coder unit No. 2). The SIF Coder Unit No. 2 shall operate as specified herein when triggered externally by pulses having the characteristics of paragraph 3.7.2.6.3. A front panel jack type UG-6253/U, or equivalent, shall accept the external trigger for coder unit No. 2."

Page 11

3.7.2.6.1.2, item (c), line 2: Delete "specifie" and substitute "specific".

3.7.2.6.5, line 3: After "than" add "2.0 µsec".

3.7.2.7.2, line 2: Delete "posible" and substitute "possible".

Page 12

3.7.2.7.4.2, line 1: Delete "haveing" and substitute "having".

Page 14

3.7.2.9.1, line 2: After "53," add "55,".

Page 15

3.7.2.11.7 Delete in its entirety and substitute:

"3.7.2.11.7 Mixed video control. A control shall be provided on the front panel to select the mixed video functions. The mixed video control shall determine what signals are available at the mixed video output jacks

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(3.7.2.12.2) only. Outputs at the SIF Coder unit No. 2 output jack (3.7.2.5.3), the challenge/tag output jack (3.7.2.12.3.3) or the reset output jacks (3.7.2.13.1) shall also be present as well as the mixed video and shall be in accordance with table I.

Table I. Mixed video control.

Switch position	Video output jacks			
	Mixed Video	SIF 2	Chal/Tag	Reset drive and reset var ampl
(a) SIF 1	SIF 1 shall appear alone. The train shall be triggered by the delay trigger.	Follows SIF 1 by the amount of the SIF 2 trigger delay (3.7.2.5.1)	Triggered by delay trigger.	Follows Chal/Tag P3 pulse by the amount of the Reset Delay (3.7.2.11.8)
(b) SIF 1 and 2	Both SIF 1 and 2 Signals shall appear as mixed video. SIF 1 shall be triggered by the delay trigger. SIF 2 shall follow SIF 1 by the amount of the SIF 2 Trigger delay (3.7.2.5.1)	Same as a.	Same as a.	Same as a.
(c) Tag-Reset	Tag and Reset signals shall appear as mixed video. Tags shall be triggered by the delay trigger. Reset shall follow the tag P3 pulse by the amount of the Reset delay (3.7.2.11.8)	Follows tag P3 pulse by the amount of the SIF delay (3.7.2.11.8) plus the SIF 2 trigger delay (3.7.2.5.1)	Same as a.	Same as a.
(d) Tag-SIF 1-Reset	The tag, SIF 1 and reset signals shall appear as mixed video. The tags shall be triggered by the delay trigger. The SIF 1 shall follow the tag P3 pulse by the amount of the SIF delay (3.7.2.11.8). The reset shall follow the tag P3 by the amount of the reset delay (3.7.2.11.8).	Same as a.	Same as a.	Same as a.

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Table I. Mixed video control (continued).

Switch position	Video output jacks			
	Mixed Video	SIF 2	Chal/Tag	Reset drive and reset var ampl
(e) All SIF	The tag, SIF 1, SIF 2 and reset signals shall all appear as mixed video. Timing shall be the same as d. with the addition of SIF 2 which shall follow SIF 1 by the amount of the SIF 2 trigger delay (3.7.2.5.1).	Same as a.	Same as a.	Same as a.
(f) Mode 4 (Word A)	Mode 4 (Word A) shall appear alone.	No output	No output	No output
(g) Mode 4 (Word B)	Mode 4 (Word B) shall appear alone.	No output	No output	No output

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3.7.3.1, last sentence: Delete.

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3.7.3.5.1, Under column headed "Markers (μsec)", delete "50.1" and substitute "50.0".

3.7.3.5.6 Delete in its entirety and substitute:

"3.7.3.5.6 Marker displacement and delay. A front panel control shall permit horizontal displacement of all markers a total of 5.0 microseconds. Sweep delay controls (see 3.7.3.2) shall simultaneously delay the crystal markers (when selected) and the display video."

Page 18

Add new paragraph as follows:

"3.7.3.7.1.1 Scale illumination. A front panel control shall be provided to illuminate the scale graticules continuously variable from fully "off" to fully "on"."

Add new paragraph as follows:

"3.7.3.7.1.2 Horizontal trace ro
provided for adjusting the alignment o
is parallel with the horizontal scale

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3.7.3.7.6, line 2: Delete "wiht" and substitute "with".

Page 20

3.7.3.7.12, line 2: After "(dB)" add "of the response at dc)".

3.7.4.1, last sentence: Delete.

Page 23

3.7.4.6.7.a Delete in its entirety and substitute:

"a. X readout gate. The X readout gate shall be adjustable from less than 150 microseconds to greater than 1200 microseconds following the input trigger. The X readout gate shall be 240 μsec in duration."

3.7.4.6.11 Delete in its entirety and substitute:

"3.7.4.6.11 Noise generator. The equipment shall provide a noise generator generating "white" noise of positive polarity in the bandwidth between 3 MHz and 10 MHz. The spectrum of the noise between 3 MHz and 10 MHz shall have no more than 3 dB difference from the frequency (or frequencies) with the highest energy content and the frequency (or frequencies) with the lowest energy content. The roll off of the spectrum shall be such that at 0 MHz it is at least 20 dB down from the maximum, at 20 MHz it is at least 0 dB down from the maximum and for all frequencies above 50 MHz the spectrum shall remain at least 20 dB below the maximum. The noise generator shall have an amplitude of at least 1.5 volts rms, when terminated by 75 ohms ±5 percent. A front panel control shall permit adjustment of the noise level from maximum to less than 0.1 volt, rms, as measured at the output jack."

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3.7.5.1, last sentence: Delete.

Page 25

Add new paragraph as follows:

"3.7.5.4.1 Internal 1030 MHz source output. Upon disconnection of the internal 1030 MHz RF source by the switch of 3.7.5.4, the internal RF shall be made available as a cw output at the module front panel. RF power output shall not be less than +25 dBm. The front panel jack shall be type UG909/U or equivalent."

3.7.5.5.2, first sentence, delete and substitute:

3.7.5.5.2 Modulation fidelity. When modulated by SIF pulse trains of

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nominal characteristics and 4.0 ±0.5 volt amplitude, the rf signal shall have the following characteristics:"

3.7.5.5.3, first sentence, delete and substitute:

"3.7.5.5.3 R.F. signal. When modulated by SIF module challenge pulses of 4.0 ±0.5 volt amplitude, the rf signal shall have the following characteristics:"

3.7.5.5.4, third sentence, delete and substitute:

"The modulation input signal shall be 0 ±0.5 volt for the fully "off" condition and no greater than 1.5 volts for the fully "on" condition. Input impedance shall not be less than 90 ohms."

Page 26

3.7.5.8.5, delete and substitute:

"3.7.5.8.5 Pulse video fidelity. Under all conditions specified herein, including the situation wherein the RF output of the Test Set is demodulated, the fidelity of the signal present at the video out jack shall be such that distortion of pulse width is not greater than ±20 nanoseconds. The pulse rise and fall time distortion shall not be greater than ±10 nanoseconds, when terminated by 75 ohms. The video output shall be provided on a front panel jack type UG-625B/U in accordance with MIL-C-39012/21."

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Add new paragraph 3.7.7.1:

"3.7.7.1 Leakage current. When practicable, the leakage current of the equipment as measured by the method specified in 4.5.6 shall not exceed 5 mA AC or DC. On equipments which have an unavoidable leakage current in excess of 5 mA, a warning plate shall be attached to the front panel reading:

'DANGER - Do not energize or connect power to this equipment unless frame and all exposed metal parts are grounded.'"

Add new paragraph 3.7.11:

"3.7.11 Panel layouts. The approval panel layouts of each module is as follows:

(a) SIF module front panel - FIGURE 2
(b) Oscilloscope module front panel - FIGURE 3
(c) Interrogator module front panel - FIGURE 4
(d) RF module front panel - FIGURE 5

Any deviation from this configuration shall be submitted to the command or agency for approval."

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4.3, last line: Add new sentence: "The leakage current test shall be in accordance with 4.5.6."

Page 30

4.5, last line: Add new sentence: "The leakage current test shall be in accordance with 4.5.6."

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Add new paragraph 4.5.6:

"4.5.6 Leakage current. Leakage current shall be measured in the following manner. Each equipment directly connected to an external power source and units deriving power from the equipment shall be placed on an insulated surface. All safety ground conductors between the equipment and units deriving power from the equipment shall be intact. The safety ground conductor between the equipment under test and the source power shall be opened. The equipment shall be connected as shown in FIGURE 6. The leakage current shall be measured in terms of voltage. The voltage measured across the 1500 ohm resistor shall not exceed 7.5 volts at the highest nominal voltage and any power line frequency for which the equipment is designed. The input impedance of the voltmeter shall be no less than one megohm. The probe shall be connected to the equipment enclosure and measurements taken for every combination of switch positions available in FIGURE 6. The probe shall also be connected to all external conducting parts such as connector housings, case, control shafts, and so forth, and the voltage measured for every combination of switch positions available in FIGURE 6. Equipments having more than one mode of operation, such as transmit and receive, shall be tested in each mode."

4.7, item (c), line 4: Delete "bureau" and substitute "command".

Page 34

FIGURE 2: Delete, and substitute new FIGURE 2.

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FIGURE 3: Delete, and substitute new FIGURE 3.

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FIGURE 4: Delete, and substitute new FIGURE 4.

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FIGURE 5: Delete, and substitute new FIGURE 5.

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Add new FIGURE 6.

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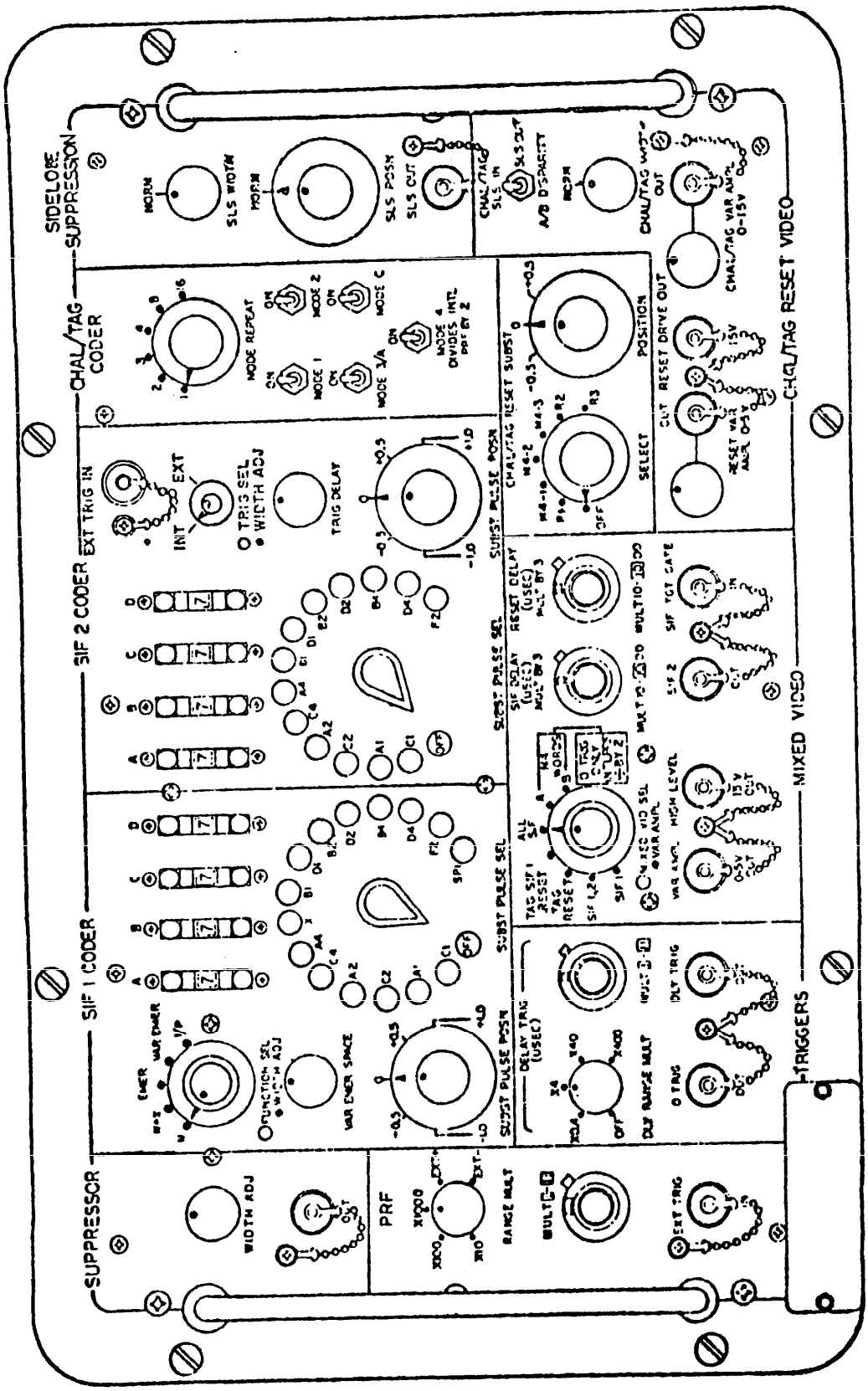


Figure 2. SELECTIVE IDENTIFICATION SIMULATOR MODULE FRONT PANEL

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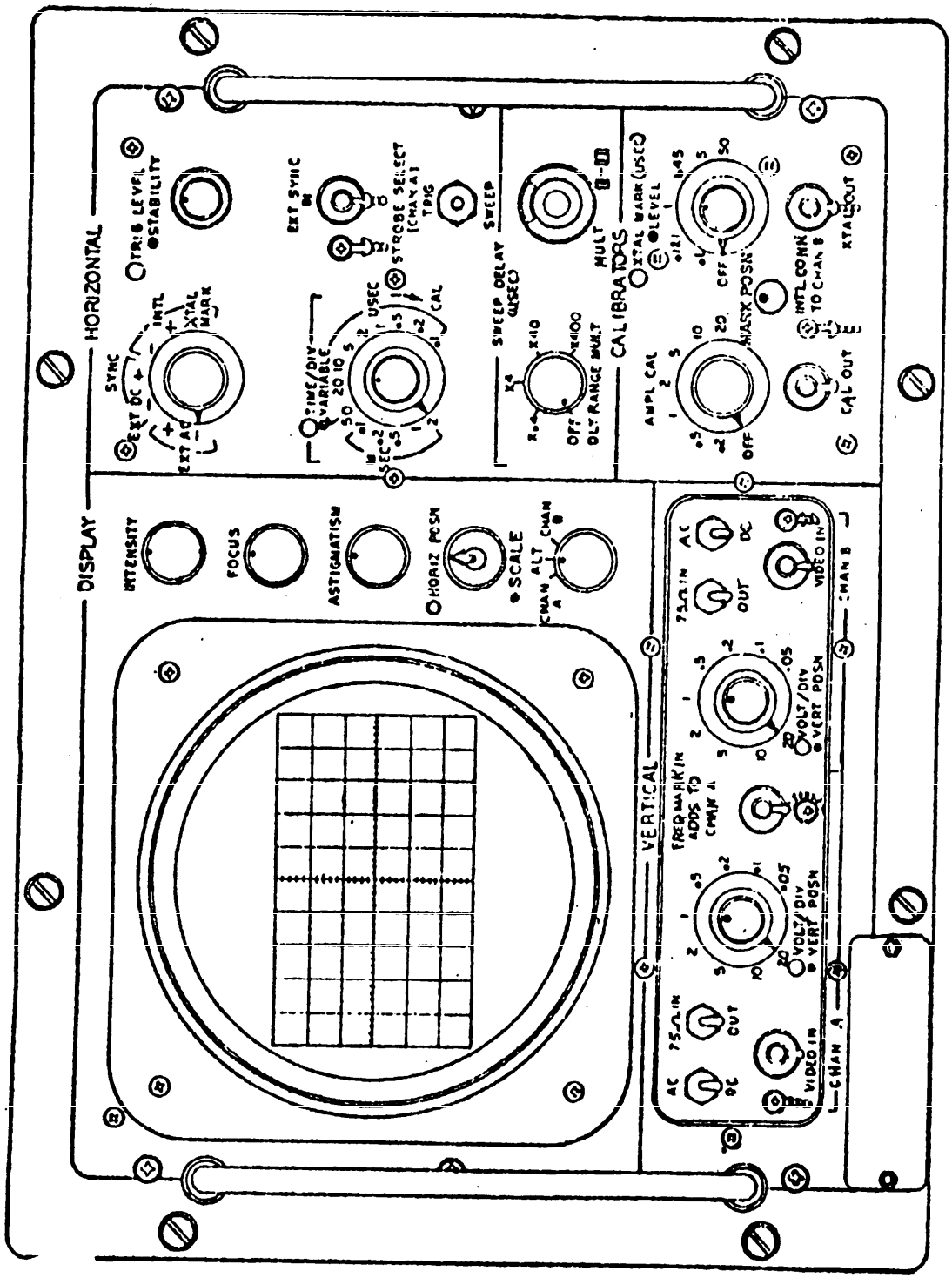


Figure 3. OSCILLOSCOPE MODULE FRONT PANEL

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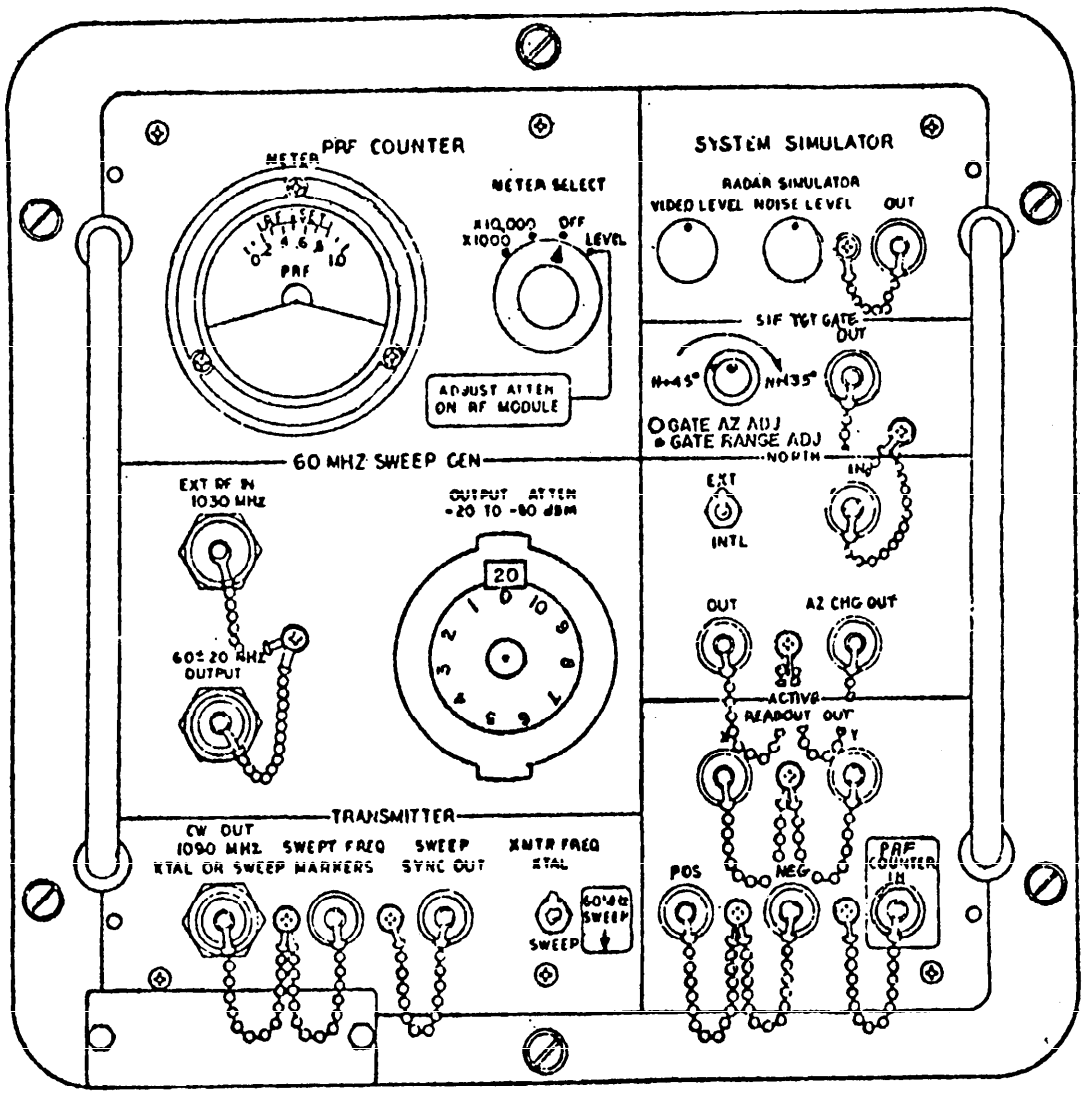


Figure 4. INTERROGATOR SIGNAL SIMULATOR MODULE FRONT PANEL

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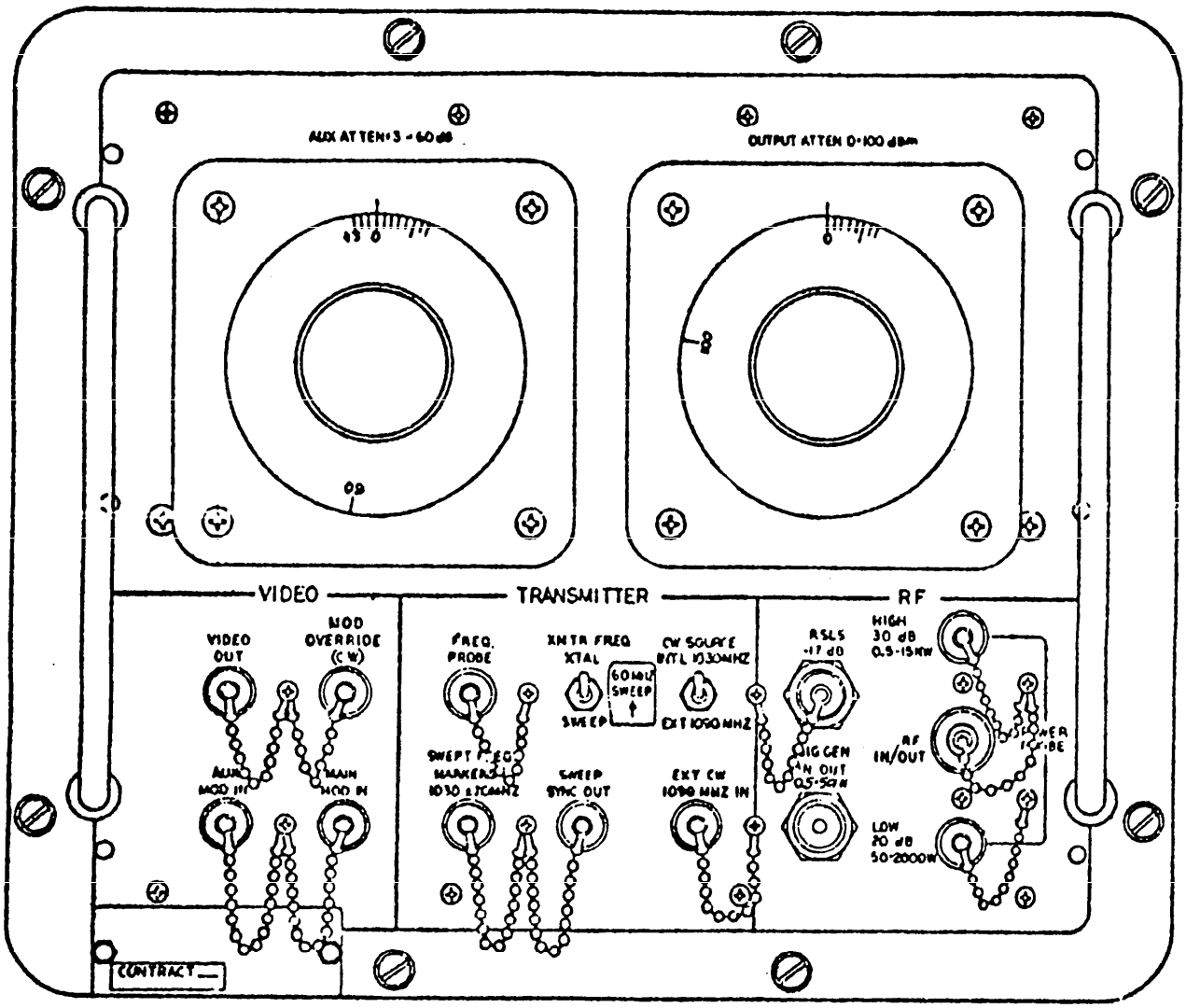


Figure 5. RF SIGNAL GENERATOR MODULE FRONT PANEL

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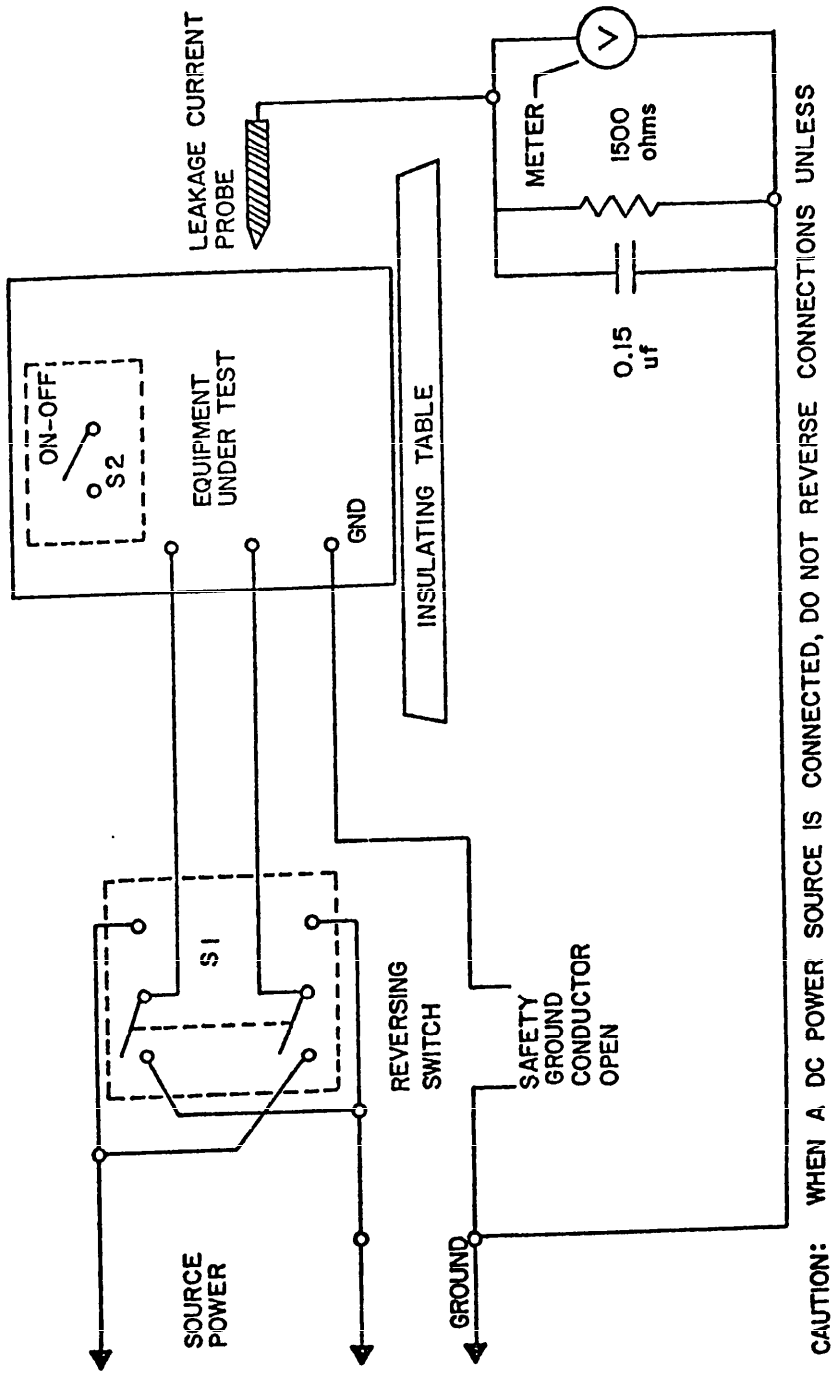


FIGURE 6 - EQUIPMENT LEAKAGE CURRENT TEST SET-UP

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